

XRPL Validator — Daily Engineering Report • 2026-09-09

Compiled retrospectively on 2026-09-23 from the git history. Scope: XRPL work only — the git history of [xrpl-validator](#) and the XRPL Rust SDK, the XRP Ledger itself (amendments) and XRPL research.

Summary

- **5 commits** (xrpl-validator 5)
- **5 engine findings fixed** (F238–F242)
- Amendment **fixCleanup3_3_0** held majority (since 2026-08-28 11:15 UTC); activation due ~2026-09-11 11:15 UTC

Engine findings fixed

- **fix(xrpl-ledger): F238 — an owner-directory insert sorts the whole page, not just the newcomer's slot** (1274170, xrpl-validator) rippled's dirAdd for an owner directory (preserveOrder == false — ApplyView.cpp directory::insertKey) sorts the last page's existing entries (std::ranges::sort(indexes)) and only then inserts the key at its lower bound. Pages written before SortedDirectories still hold their entries in insertion order until something is added to them, and the next insert then sorts every entry, not only the new one. Our dir_insert ...
- **fix(xrpl-ledger): F239 — a bridged strand's rev pass reaps past the level it consumes whole** (0ed3d96, xrpl-validator) The direct walk has run rev_extent_reap per level since F114: rippled's rev pass steps a book by the strand's WANT, not by the taker's funds, so it consumes the head level whole and then steps — reaping every dead offer it passes until a live one on a worse level ends the iteration (*ofrQ != offer.quality(), BookStep.cpp:719). cross_bridged never did. Its only removals came from the F233 gate, which reaps where a ...
- **fix(xrpl-ledger): F240 — an XRP amount of more than sixteen digits is exact, in what rests and in the rate it rests at** (5d1a89d, xrpl-validator) A native STAmount carries drops as an exact integer up to 1e17. It is not held to the sixteen-digit mantissa an IOU is canonicalized into, and two places in the resting path assumed otherwise.
- **fix(xrpl-ledger): F241 — a check writer's shortfall is refused in preclaim, as tecPATH_PARTIAL** (e37e205, xrpl-validator) rippled checks the writer's balance before the payment engine ever runs (CheckCash.cpp:169-192): availableFunds = accountFunds(check.Account, value, ZeroIfFrozen, ZeroIfUnauthorized) if (value.native() && !check[sfSponsor]) availableFunds += fees().increment if (value > availableFunds) return tecPATH_PARTIAL
- **fix(xrpl-ledger): F242 — unauthorizing a DepositPreauth drops the owner directory it empties** (3fefb0d, xrpl-validator) rippled unlinks a DepositPreauth with keepRoot FALSE: view.dirRemove(keylet::ownerDir(account), page, preauthIndex, false) DepositPreauth.cpp:283

Amendment watch

Read from the XRP Ledger: the Amendments object as of the last ledger of the day, and the day's EnableAmendment pseudo-transactions.

Amendment	Majority since	Activation due
fixCleanup3_3_0	2026-08-28 11:15 UTC	2026-09-11 11:15 UTC

Amendments enabled on mainnet: 93.

XRPL research

Automated research — candidate amendment ideas

Ideas generated and prior-art-checked by the XRPL Amendment Board, an automated research tool (read-only board: iceland.halcyon-names.io). They are exploration leads, not proposals.

- **Confidential Vault Share Balances** (lending) Add an lsfVaultConfidential flag to the Vault ledger object (SingleAssetVault); when set, VaultDeposit/VaultWithdraw store each depositor's position a
- **BLS Light-Client Verified Bridge Headers** (interop) Add an owner-reserved ledger object ltXCHAIN_LIGHT_CLIENT (keylet derived from the Bridge object plus door account) holding a counterparty chain's tru
- **Credential-Bound Session Key Revocation** (accounts) Extend g11's SessionKey ledger object with optional SFields sfCredentialType and sfCredentialIssuer, settable only at creation via SessionKeySet. When

Appendix — commits

Time	Repo	Commit	Subject
01:24	xrpl-validator	1274170	fix(xrpl-ledger): F238 — an owner-directory insert sorts the whole page, not just the newcomer's slot
07:52	xrpl-validator	0ed3d96	fix(xrpl-ledger): F239 — a bridged strand's rev pass reaps past the level it consumes whole
09:22	xrpl-validator	5d1a89d	fix(xrpl-ledger): F240 — an XRP amount of more than sixteen digits is exact, in what rests and in the rate it rests at
10:08	xrpl-validator	e37e205	fix(xrpl-ledger): F241 — a check writer's shortfall is refused in preclaim, as tecPATH_PARTIAL
11:03	xrpl-validator	3fefb0d	fix(xrpl-ledger): F242 — unauthorizing a DepositPreauth drops the owner directory it empties